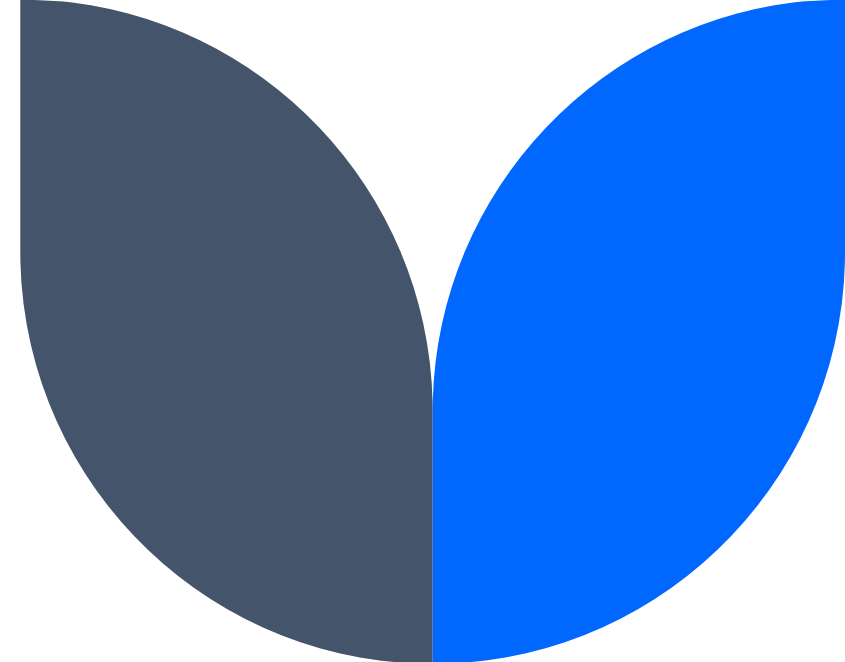
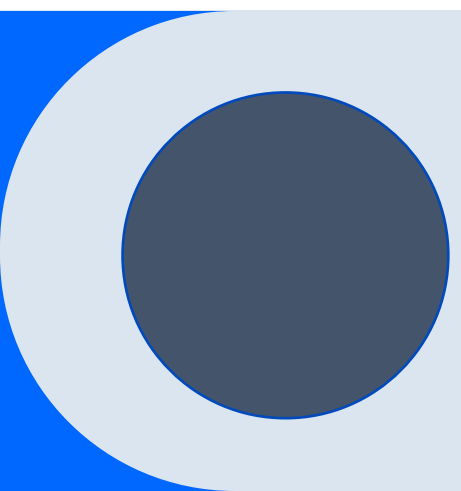




Customer Churn Classification Using Machine Learning



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Project Overview

My final project focused on predicting customer churn for a telecommunications company. Customer churn occurs when a customer stops using a company's services. I used a dataset containing 7,043 customer records and information about customer accounts, contracts, services, and billing. The target variable was Churn, which contained two categories: Yes and No. The main goal was to compare several machine-learning models and determine which model could most effectively identify customers who were likely to leave.

Methodology

Data Preparation

The dataset contained 7,043 customer records. During cleaning, the TotalCharges column was converted from text to numerical values, and 11 missing entries were replaced with zero because those customers had zero months of tenure. The customerID column was removed because it was only an identifier and did not provide useful information for predicting churn.

Feature Engineering and Data Split

Two new features were created. TenureGroup organized customers based on how long they had remained with the company, while NumberOfServices counted the services used by each customer. The completed dataset was divided into 70% training data, 15% validation data, and 15% testing data using stratification.

Model Development

Numerical features were standardized, categorical features were one-hot encoded, and RandomOverSampler was applied only to the training data to address class imbalance. Six classification models were trained and compared: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, K-Nearest Neighbors, and Support Vector Classifier. An Average Ensemble and a Bayesian Ensemble were also created using the strongest individual models.

Process Flow:

Data Cleaning → Feature Engineering → 70/15/15 Split → Preprocessing → Class Balancing → Model Training → Validation Comparison → Final Test Evaluation



Model Results

Gradient Boosting was the strongest individual validation model, producing a ROC-AUC of 0.8376 and an F1-score of 0.6369. The Average Ensemble and Bayesian Ensemble performed similarly, showing that the strongest individual models produced consistent predictions.

Final Test Performance

The Bayesian Ensemble achieved the strongest overall test result. It combined predictions from Gradient Boosting, Random Forest, and Logistic Regression.

Metric	Bayesian Ensemble
Accuracy	0.7711
Precision	0.5470
Recall	0.7893
F1-Score	0.6462
ROC-AUC	0.8531

Key Finding

The Bayesian Ensemble correctly identified approximately 79% of customers who churned. It produced the highest test ROC-AUC and the best balance between accuracy, precision, recall, and F1-score.

Best Individual Model: Gradient BoostingBest Overall Model: Bayesian Ensemble

Weakest Model: K-Nearest Neighbors

Conclusion and Reflection

Conclusion

This project showed that machine-learning models can use customer account, service, contract, and billing information to identify customers who are likely to churn. The Bayesian Ensemble produced the strongest final performance, with a test ROC-AUC of **0.8531** and recall of **0.7893**. Its high recall means it identified most customers who actually left the company.

Limitations and Improvements

The project used selected model settings rather than a full hyperparameter search, and RandomOverSampler balanced the training data by duplicating minority-class records. Future improvements could include hyperparameter tuning, SMOTE, probability calibration, and testing different classification thresholds based on the cost of customer-retention efforts.

Reflection

This project helped me understand the importance of cleaning data before modeling, preventing data leakage, and using separate training, validation, and test sets. I also learned that accuracy alone is not enough for an imbalanced classification problem. Comparing precision, recall, F1-score, and ROC-AUC provided a clearer understanding of each model's performance.

Final Recommendation

Use the Bayesian Ensemble to rank customers by churn risk and support targeted retention decisions.